

CPSC 317 – Summer 2026

Introduction to Computer Networking

# **CPSC 317: Introduction to Computer Networking**

## **Module 0: Introduction and Internet Design**

Ibtissem

# Things to do today

- Introduce ourselves
- Talk a little about the course content
- Talk a little more about the course structure and rules

# Who am I?

- Ibtissem Bouacheria
  - Sessional Lecturer, Department of Computer Science
  - 5+ years teaching experience
- Office: ICCS 249
- My interests: networking, Internet Of Things, routing protocols

# Who am I?

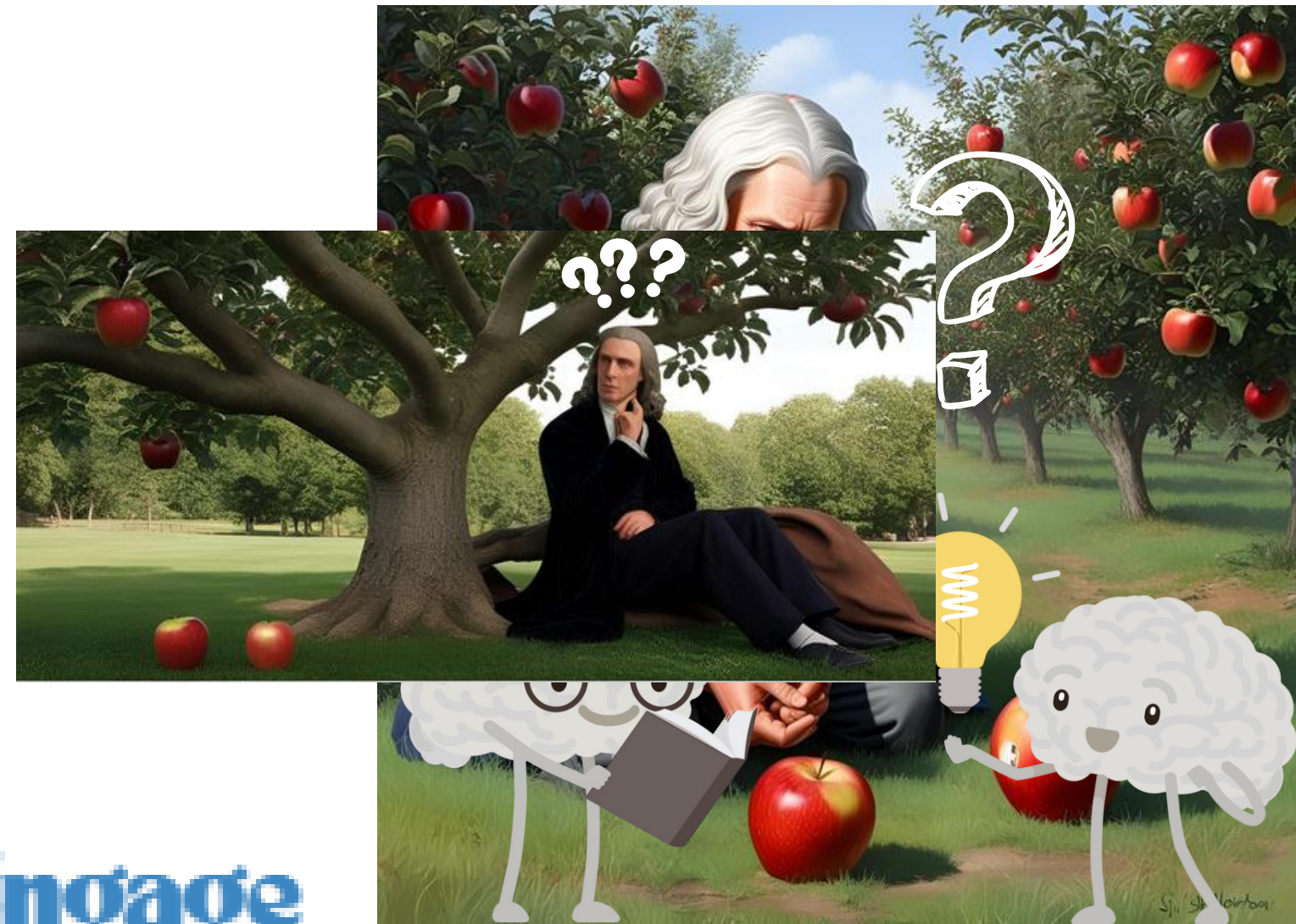
Quote that shaped my life

*“Knowledge can be taken, but not given”*



# What do you think?

*“Knowledge can be taken, but not given”*



**Stay Curious ! Ask Questions ? Fully Engage & Learn**

# Introduce Yourself

- Talk to your neighbours
- Introduce yourself to each other

## Topics of discussion

- What area of Computer Science are you interested in?
- Why are you taking this course?
  - What do you want to learn?



# Pre-Requisites

- CPSC 213 (Introduction to Computer Systems)
- CPSC 221 (Basic Algorithms and Data Structures)
- If you don't have these prerequisites, check
  - <https://www.cs.ubc.ca/prerequisite-rules>

# Course Information

## ❖Lecture:

- Instructor: Ibtissem Bouacheria
- Section 911
  - Class time : Tue/Thu at 11:00AM – 2:30PM



# Course Information

❖ **Tutorials:** Sections D1A/D1B/D1C/D1D

– Tutorial time :

- D1A: Tue/Thu at 03:30PM – 04:30PM
- D1B: Tue/Thu at 04:30PM – 05:30PM
- D1C: Tue/Thu at 05:30PM – 06:30PM
- D1D: Tue/Thu at 06:30PM – 07:30PM

# Tutorials

- **What will be covered in tutorials?**
  - Practical exercises based on course content and programming assignments.
- **How many tutorials are there?**
  - 10 tutorials in total.
- **When do tutorials start?**
  - They begin this week.

# Tutorial Registration

- **Do I have to go to a tutorial?**
  - Yes
- **I want a tutorial, but sections don't fit my schedule**
  - Keep looking – there is often quite a bit of “churn” in the first weeks
  - We'll keep an eye on registration numbers
  - Attend other tutorials in the meantime

# Course Tools



**Canvas**

**A**nnouncements

**S**yllabus

**T**extbook

**A**dditional references

**L**inks to all other tools



Canvas

Announcements

Syllabus

**Textbook**

Additional references

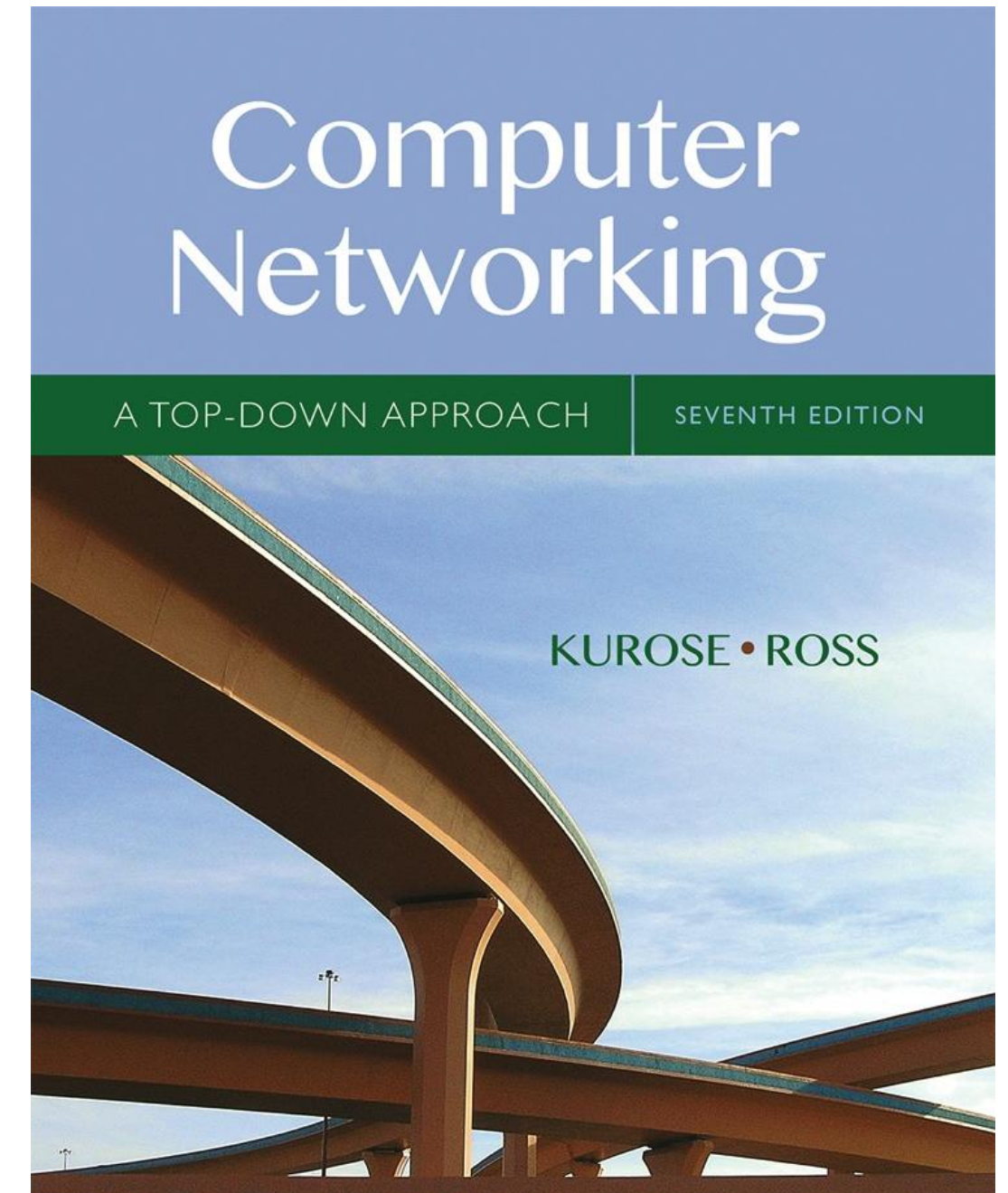
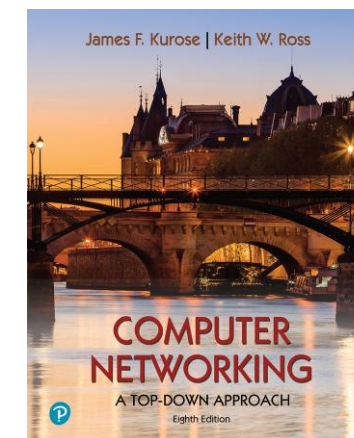
Links to all other tools

## Textbook

*Computer Networking: A Top-Down Approach,*

7th edition,  
Jim Kurose, Keith Ross,  
Pearson, 2017

Use 8th edition if you're planning  
to take CPSC 417 in the future





# Course Tools



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iClicker

In-class

Participation

Questions

( 3% Bonus Grade)





In-class  
Participation  
Questions

( 3% Bonus Grade)

## iClicker Cloud

- In-class participation questions
  - Typically, multiple-choice or short answer questions
- 3% Bonus Grade
- Answering gets you half the points per question
  - Answering correctly gets you the other half
- Please use a valid student ID when logging in
- Section 911: Join: <https://join.iclicker.com/MRYF>

# Course Tools



**Canvas**

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**T**extbook

**A**dditional references

**L**inks to all other tools

( **3%** Bonus Grade)



**iClicker**

In-class

Participation

Questions

**PrairieLearn**

**PrairieLearn**

**Practice Part**

In-Class Activities (ICA)

Quizzes (PFQ)

**Graded Part**

Assignments (**35%**)

Quizzes (**30%**)

Final Exam (**35%**)

**Request a review**

PrairieLearn

PrairieLearn

Practice Part

In-Class Activities (ICA)  
Quizzes (PFQ)

Graded Part

**Assignments (35%)**

Quizzes (30%)  
Final Exam (35%)

Request a review

- **Programming Assignments** (35% of your final grade)
  - Will be released and **submitted on PrairieLearn**
  - You will have **five assignments**
    - PA1: Implementing a DICT client
    - PA2: Implementing a DNS client
    - PA4: Implementing a reliable transport protocol
    - PA5: Implementing forwarding in a router
  - **Submissions** may have **preliminary autograding on PrairieLearn**
    - **TAs will still review** all submissions
  - All **assignments** to be done **individually**, except PA4

PrairieLearn

## PrairieLearn

### Practice Part

In-Class Activities (ICA)  
Quizzes (PFQ)

### Graded Part

Assignments (35%)

**Quizzes (30%)**

**Final Exam (35%)**

### Request a review

## Quizzes and Final Exam Format

- Short answer questions / Multiple-choice questions
- Computer-Based Testing Facility (CBTF)
- Synchronous
- **Quizzes** (30% of your final grade)
  - Weekly quizzes
  - In CBTF, throughout the week
  - First quiz self-scheduled from May 20<sup>th</sup> to May 23<sup>rd</sup>.

## Final Exam (35% of your final grade)

- Do not schedule travel in April until you know when the exam is happening (and even then, be conservative – unforeseen events (fire alarms, etc.) can cause the final exam schedule to change at the last minute)

# Course Tools



**Canvas**

Announcements

Syllabus

Textbook

Additional references

Links to all other tools

( 3% Bonus Grade)



**iClicker**

In-class  
Participation  
Questions

**PrairieLearn**

**PrairieLearn**

**Practice Part**

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Quizzes (PFQ)

**Graded Part**

Assignments (35%)  
Quizzes (30%)  
Final Exam (35%)

**Request a review**



**GitHub**

Slides





## Slides

### Slides

- Slides will be regularly posted on GitHub and Canvas
- As much as possible will be posted in advance
  - Sometimes they may be modified shortly before class

<https://github.students.cs.ubc.ca/cpsc317/Slides-2026S1>



# Course Tools



**Canvas**

Announcements

Syllabus

Textbook

Lecture Slides

Additional references

Links to all other tools (3% Bonus Grade)



**iClicker**

In-class

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Practice Part

In-Class Activities (ICA)

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**GitHub**

Slides



**Piazza**

Discussion board



Piazza

Discussion board

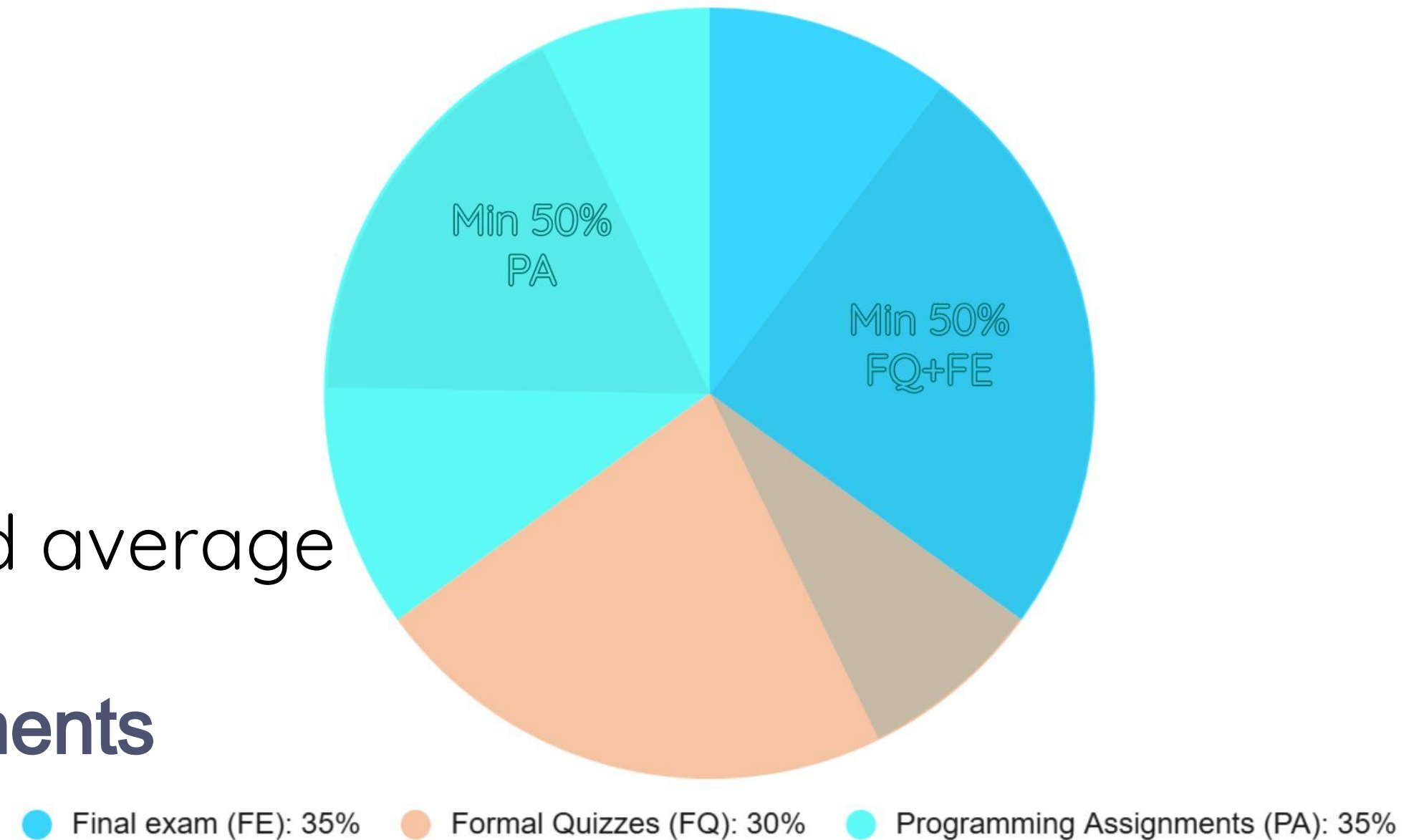
## Piazza

- Discussion board
  - Do not use the discussion board on Canvas
- To join:
  - Use the link below. It is provided on Canvas (or search Piazza for this course)

[https://piazza.com/ubc.ca/summer2026/cpsc\\_v3179112026ss](https://piazza.com/ubc.ca/summer2026/cpsc_v3179112026ss)

## Grading Scheme

- **Programming Assignments: 35%**
- **Quizzes: 30%**
- **Final Exam: 35%**
- You **must** get **50%** in the weighted average of the **quizzes** and the **final exam**.
- You **must** get **50%** on the **assignments**
- **Participation:**
  - Lectures: up to **3% Bonus** (based on active presence and iClicker questions in class)
  - Tutorials: up to **2% Bonus** (based on active presence)
- We reserve the right to make minor modifications to this scheme



# Contact Information

- For **most requests email**: [cpsc317-admin@cs.ubc.ca](mailto:cpsc317-admin@cs.ubc.ca)
  - Extensions, Illness accommodations, ...
- To **contact the instructor only**: [ibtissem.bouacheria@ubc.ca](mailto:ibtissem.bouacheria@ubc.ca)
- **Our teaching assistants** are:
  - Han, Kevin, Kyle and Satvik
- **Office hours**: To be announced on Piazza
- For **general questions** about **course material or assignments**, **do not email** directly. Instead:
  - Use **Piazza**
  - Ask a TA during **office hours**
  - Ask instructor **before or after class** or during office hours

## Some Rules: Absences

- If you must be absent in a quiz, contact the course coordinator coordinator ([cpsc317-admin@cs.ubc.ca](mailto:cpsc317-admin@cs.ubc.ca)) as soon as you are aware of the problem
- You do not need to notify us of absences in days when there is no graded work
- For assignments, there is no special consideration for expected absences
- Full policy and notification instructions available on course syllabus page.



## Some Rules: Marking Disagreements

- You have 1 week from when a piece of graded work is handed back to request a review
  - Review requests must be submitted through PrairieLearn as reported issues
  - After 1 week the mark stands
- We may review marking of other questions
- Full policy and notification instructions available on Course Syllabus

# Academic Conduct

- Don't cheat!!
- Review University and course policies on academic misconduct, as well as any relevant assessment instructions
- If you are uncertain: ASK!!!
- We can and will compare your solutions to those of other students (past and present)
- Possible penalties
  - Failing grade in the work in question or in the course in general
  - Suspension from the University
  - Reprimand with letter in student's file
  - Notation in student's permanent record
- Full policy and information on course syllabus



# Things To Do

- Get familiar with course tools (if you are not already):



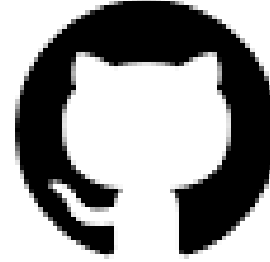
Canvas

<https://canvas.ubc.ca/courses/186807>



iClicker

<https://join.iclicker.com/MRYF>



GitHub

<https://github.students.cs.ubc.ca/cpsc317/Slides-2026S1>



PrairieLearn

[https://us.prairielearn.com/pl/course\\_instance/215049](https://us.prairielearn.com/pl/course_instance/215049)



Piazza

[https://piazza.com/ubc.ca/summer2026/cpsc\\_v3179112026ss](https://piazza.com/ubc.ca/summer2026/cpsc_v3179112026ss)

# Iclicker

<https://join.iclicker.com/MRYF>

*What was your favourite course?*  
*If you prefer not to answer, please type 'CPSC'*

# Clicker Question

- Based on your previous CPSC courses, how do computers represent and handle different types of information such as text, images, numbers, and sound?
  - A. In binary (0s and 1s).
  - B. Text and numerical data are represented in bits, while multimedia content (audio, images, video) are represented in human-readable form.
  - C. Computers use artificial intelligence to interpret data instead of binary encoding.
  - D. Data is represented directly in human-readable form.

## Iclicker Answer

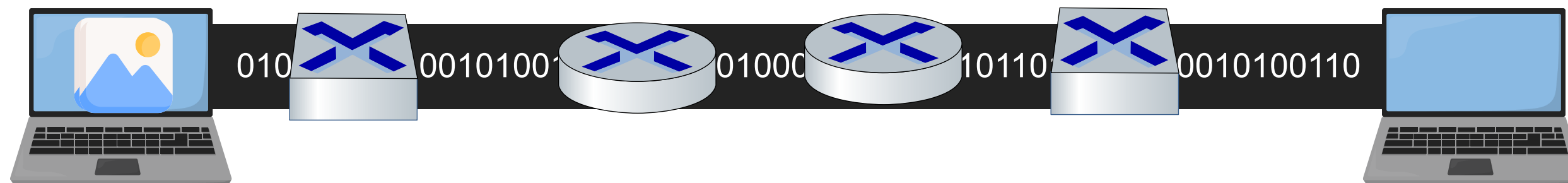
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# Reflexion

- How do we make it work?
  - Communication media (wires, radio signals, etc.)
  - Network architecture (hosts, NICs, switches, routers, etc.)
  - Protocols
  - Applications



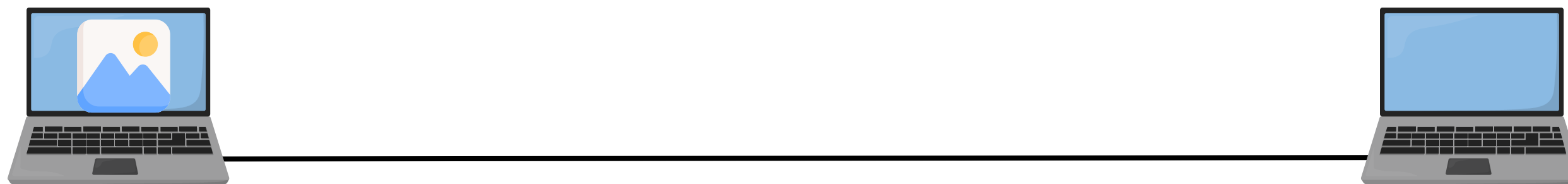
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# Reflexion

- **What is a computer network used for?**
  - Communication
  - Information sharing
  - Task distribution
  - Resource sharing
  - Scaling and redundancy



# Coping with complexity

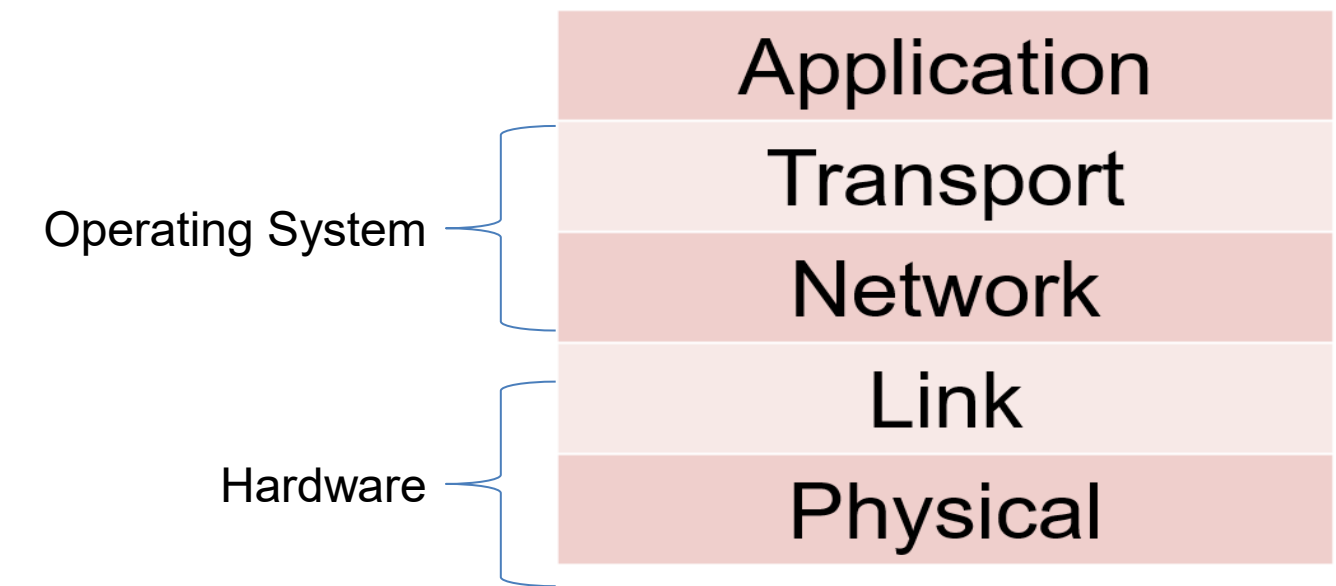
- **Complexity** limits the size of the systems we build
  - We need techniques to cope with it in order to build larger systems
- **Modularity**: Divide system into smaller components
- **Abstraction**: Define how a component interacts with the outside world. Then don't worry about it works internally





# Coping with complexity (cont.)

- **Robustness drives independence**
  - Be tolerant of inputs (deal with incorrect values sensibly)
  - Be strict on outputs (don't produce unexpected values)
- **Layering**
  - A particularly successful type of modularity
  - The system contains a number of levels, each built on the previous one
  - Used in networks and operating systems
- **Hierarchy**
  - A more general approach when layering is too strict
- **Indirection**
  - Names are the glue that holds systems together
  - Using indirection allows sub-systems to be swapped more easily



# Why Study Computer Networking?

- Unlike most courses the focus is on **COMMUNICATION** not computation.
- What is different? Communication requires at least two parties.
- Plumbing -- what underlies the applications in wide use today.
- The applications enabled by networking are the most disruptive technology over the past 30 years

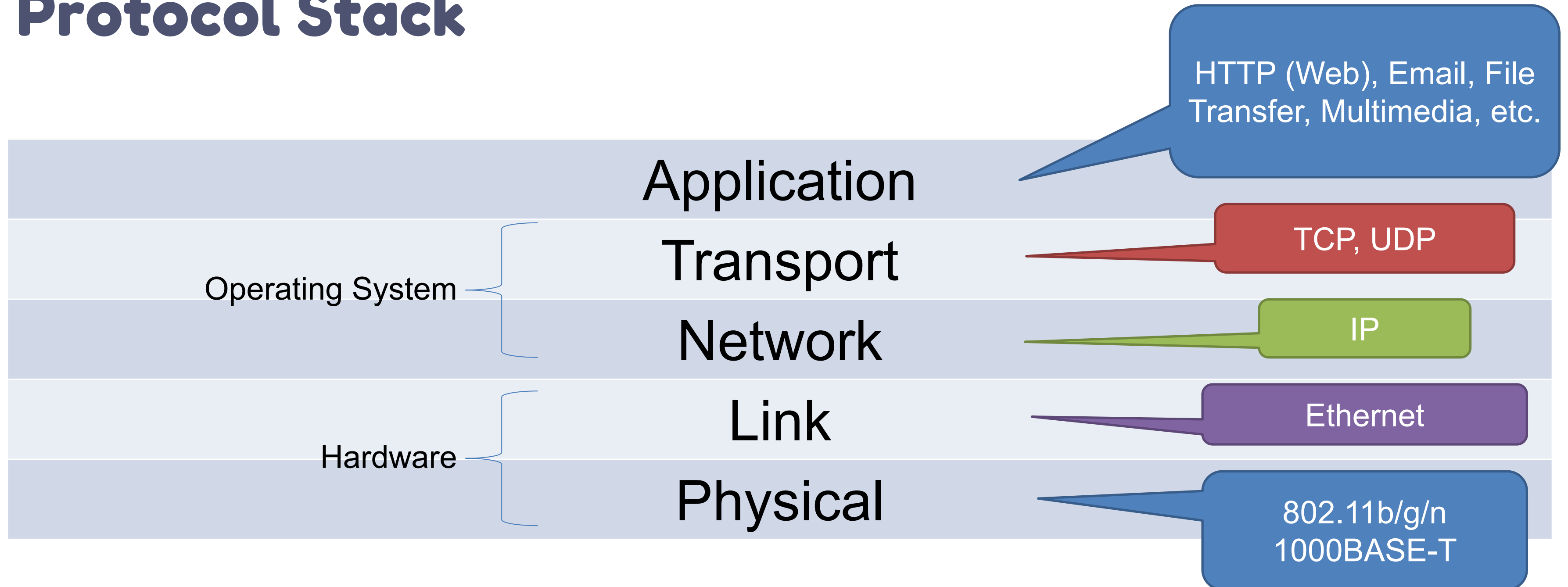
# Course Objectives

- You should become comfortable with:
  - writing and working with different types of programs that use computer networks
  - terminology about networking

# Course Objectives

- Key paradigms and strategies used in developing applications that use networks:
  - Isolation and privacy
  - Data loss
  - Performance
  - Naming and location
  - Using layers and abstractions
- Applying these strategies and paradigms
- Basic concepts on how the Internet is put together and works

# Protocol Stack





# **Next Topic: Design of the Internet**